Highlights

Beyond Accuracy: Explainable Trustworthy Index and Explainable Robustness Index for Evaluating CNN

Ragib Mehedi, Azmain Inquaid Haque, Md. Nazmul Abdal, Abdullah Al Noman, Md. Zahidul Islam

- Proposed two novel metrics, ETI and ERI, to assess CNN trustworthiness and robustness
- Introduced LMOS, a Grad-CAM-based score linking model focus to ground-truth object regions
- Evaluated 8 CNN architectures on 4 leaf/flower datasets under 9 real-world perturbations
- High accuracy does not guarantee reliability: Custom CNN scored 99.45% but had the lowest ETI (0.59) on EPLI
- Xception showed the most consistent explainable robustness across all datasets

Beyond Accuracy: Explainable Trustworthy Index and Explainable Robustness Index for Evaluating CNN

Ragib Mehedi[a,*], Azmain Inquaid Haque[a], Md. Nazmul Abdal[b], Abdullah Al Noman[a] and Md. Zahidul Islam[a]

[a]Computer Science and Engineering Discipline, Khulna University, Khulna, 9208, Bangladesh
[b]Department of Computer Science and Engineering, University of Liberal Arts Bangladesh, Dhaka, 1207, Bangladesh

ARTICLE INFO

Keywords:
Convolutional neural networks, Explainable AI (XAI), Trustworthiness, Robustness, Grad-CAM, Perturbation Analysis

ABSTRACT

Convolutional neural networks (CNNs) can achieve high classification accuracy while grounding their predictions on background or irrelevant image regions. Furthermore, external input perturbations can alter both what the models predict and what evidence they consider for making the predictions. Accuracy-based evaluation captures neither problem. We propose two complementary model-level metrics that integrate predictive performance with explanation quality. The Explainable Trustworthy Index (ETI) combines classification accuracy with the Localization Mask Overlap Score (LMOS). LMOS is the proportion of salient Grad-CAM activation that falls within the ground-truth object mask. The Explainable Robustness Index (ERI) measures how well a model preserves both accuracy and LMOS under input perturbations, relative to its behavior on clean images. We evaluate both metrics on four real-world leaf and flower image datasets, using eight CNN architectures and nine perturbations, including noise, blur, and occlusion. The results show that accuracy and explanation quality frequently diverge. On the EPLI dataset, a Custom CNN attains 99.45% accuracy, yet only 19% of its salient activation falls on the leaf, giving it the lowest ETI (0.5922). Across all the datasets, DenseNet121 achieves the highest accuracy but the highest ETI on none, while Xception obtains the highest ERI on all four. Predictive and explanation stability also diverge: under contrast perturbation, the Custom CNN largely preserves its accuracy while its object focus shifts substantially, whereas ResNet50 shows the opposite pattern. ETI and ERI thus expose differences in reliability and robustness that accuracy alone conceals, providing complementary criteria for model evaluation and selection.

1. Introduction

Convolutional Neural Networks (CNNs) have achieved remarkable performance in a wide range of domains such as analysis of medical imaging, agriculture, and astronomy, where predictions inform high-stakes decision-making processes [1, 2, 3, 4]. In such applications, classification accuracy is an essential performance measure, but accuracy alone does not necessarily indicate whether a model is reliable. A CNN model is considered more reliable when its predictions are grounded in task-relevant visual features rather than spurious correlations [5]. Therefore, beyond predictive accuracy, models must be quantitatively evaluated to determine whether their predictions depend on task-relevant regions rather than irrelevant background cues. Evaluating CNN models therefore requires asking not only whether its predictions are correct, but whether they are grounded on the target object or on the background around it.

Model behavior can shift due to real-world perturbations e.g., *noise*, *blur*, *lighting shifts*, *compression*, and *occlusion*. These distortions can alter both CNN accuracy and the visual regions driving predictions. Even when a model maintains relatively high accuracy under perturbations, its underlying visual explanations may still become inconsistent [5, 6]. Conventional robustness metrics are primarily based on accuracy and these may change under perturbations. They do not capture whether the model maintains consistent reliance on task-relevant regions under perturbed conditions.

Explainable Artificial Intelligence (XAI) provides a framework for examining visual evidences used by CNNs for their predictions. In particular, Grad-CAM generates class-discriminative heatmaps that highlight image regions contributing to a model's predictions [7, 8]. Consequently, Grad-CAM may show whether a model rely mainly on background and irrelevant regions instead of focused objects [9, 10]. We refer to the Grad-CAM heatmaps as explanations corresponding to a model's predictions. Previous studies have shown that background information and spurious correlations can influence CNN decisions and may lead to shortcut learning [11, 12, 13].

Prior literature has presented diverse methodologies for reliability, robustness, and explanation quality of CNNs. More specifically, reliability has been assessed via uncertainty estimation, calibration confidence scoring, failure detection, and risk-coverage trade-offs [14, 15]. Robustness is

*Corresponding author
ragib6627@gmail.com (R. Mehedi); azmaininquaidhaque@gmail.com (A.I. Haque); nazmul.abdal@ulab.edu.bd (Md.N. Abdal); noman@cse.ku.ac.bd (A.A. Noman); zahid@cse.ku.ac.bd (Md.Z. Islam)
https://ulab.edu.bd/faculty/md-nazmul-abdal (Md.N. Abdal); https://ku.ac.bd/discipline/cse/faculty/noman (A.A. Noman); https://ku.ac.bd/discipline/cse/faculty/zahid (Md.Z. Islam)
ORCID(s): https://orcid.org/0009-0003-1555-7338 (R. Mehedi); https://orcid.org/0009-0009-4032-2622 (A.I. Haque); https://orcid.org/0009-0001-5233-0920 (Md.N. Abdal); https://orcid.org/0000-0001-7537-1402 (A.A. Noman); https://orcid.org/0000-0002-9209-9524 (Md.Z. Islam)

typically evaluated by measuring performance degradation under common corruptions [16, 17], natural distribution shifts [18], and targeted input perturbations [19, 20]. Recent efforts also examined how corruptions affect explanation stability [21]. Explanation quality and visual feature reliance have been measured using faithfulness metrics, ground-truth alignment, and the spatial overlap between visual explanations and target object regions [22, 23, 24]. However, such localization measures are typically used to compare explanation methods on a fixed model, and are reported separately from accuracy rather than combined with it into a model-level score. Similarly, explanation-stability analyses [21] quantify how much a heatmap changes under corruption, but not whether it remains on the target object, and treat this independently of changes in accuracy. Despite these advances, existing methods lack a unified, model-level evaluation framework that simultaneously accounts for predictive accuracy, the spatial fidelity of decision-relevant visual evidence, and their joint stability under input perturbations.

We propose two novel evaluation metrics: *Explainable Trustworthy Index (ETI)* and *Explainable Robustness Index (ERI)* using XAI methods. Our objective is not to improve CNN models, rather to provide quantitative measures for evaluating models and comparing their reliability and robustness. ETI measures baseline reliability on unperturbed images by jointly evaluating classification accuracy and object-focused explanation quality. In contrast, ERI measures robustness to input perturbations by assessing whether accuracy and explanation quality remain consistent with clean-image behavior. In essence, ETI and ERI address two different questions: ETI asks "Whether a model is accurate and appropriately focused on the target object under normal conditions?", whereas ERI asks "Whether these properties remain stable when the input is perturbed?"

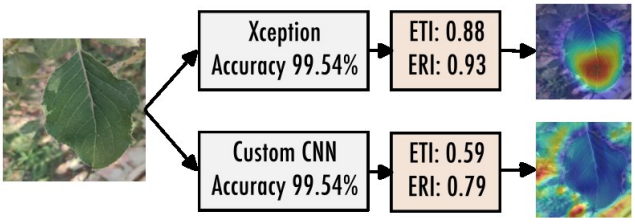


Figure 1: Accuracy versus the proposed metrics for two models trained on the EPLI dataset. Xception and the Custom CNN achieve identical accuracy (99.45%), but their Grad-CAM heatmaps for the same test image (right) differ: Xception's activation is concentrated on the leaf, whereas the Custom CNN's falls largely on the background. The proposed ETI and ERI reflect this difference.

Figure 1 illustrates the distinct advantages of our XAI-based metrics over traditional accuracy. Xception [25] and the Custom CNN (detailed in Section 5.3), trained from identical data, both achieve 99.45% accuracy on EPLI dataset [26], yet Xception obtains substantially higher ETI and ERI scores (0.88 and 0.93, versus 0.59 and 0.79). The Grad-CAM heatmaps show that the Custom CNN relies heavily on the background rather than the leaf. Accuracy alone rates the two models as equivalent, whereas ETI and ERI identify the Custom CNN as the weaker choice for deployment.

We introduce a Localization Mask Overlap Score (LMOS) to quantify object-focused explanation quality. LMOS indicates the proportion of a model's decision-relevant activation that overlaps with the annotated target-object region. LMOS is used in both ETI and ERI calculation. It is observed that in Grad-CAM visualizations, red and yellow regions are activated most i.e., red and yellow have greater influence than other colors on models' predictions [27, 28]. We integrate this principle into LMOS; a higher LMOS value indicates that a larger proportion of the model's highlighted regions falls within the target object boundary rather than the background.

We comprehensively evaluate the proposed metrics using four real-world datasets comprising two leaf image datasets: EPLI [26], BDML[29] and two flower image datasets: FLRS [30], FR [31]. Eight CNN architectures are trained and evaluated on the datasets. The gap illustrated in Figure 1 is substantial: on EPLI, only 19% of the Custom CNN's salient activation falls within the leaf, against 77.65% for MobileNetV2, despite identical accuracy. This pattern holds across datasets: DenseNet121 achieves the highest accuracy on all four datasets yet the highest ETI on none, while Xception obtains the highest ERI on all four datasets. A low LMOS does not by itself establish shortcut learning or overfitting; rather, it quantifies how far a model's salient activation diverges from the target object.

The proposed ERI combines *accuracy stability* and *LMOS stability* to provide a joint measure of explainable robustness. This distinction is important because predictive stability and explanation stability are not necessarily aligned. For example, in our experiments we observe that the Custom CNN model under contrast perturbation maintains relatively high accuracy stability of 0.8494 but has substantially lower LMOS stability of 0.4545 on the EPLI dataset. This indicates that, although its predictive performance remains comparatively stable, its object-focused explanation changes substantially relative to the clean-image condition. Conversely, the ResNet50 model exhibits relatively low accuracy stability of 0.4599 but higher LMOS stability of 0.7830 under the same perturbation, indicating that its object-focused explanation remains comparatively more consistent despite a larger change in predictive performance. These demonstrate that accuracy stability alone cannot fully characterize robustness: a model may preserve its predictive performance while substantially changing its object-focused behavior, or it may preserve its visual focus while its predictive performance deteriorates.

The significance of ETI and ERI lies in their ability to capture key dimensions of CNN reliability and robustness that standard accuracy metrics overlook. However, a high ETI or ERI score should not be equated with overall model trustworthiness, which encompasses broader factors beyond the scope of this work. Rather, these metrics focus

specifically on predictive accuracy, object-focused explanation quality, and their stability under input perturbations - offering complementary criteria for evaluating CNN models beyond conventional accuracy benchmarks.

2. Problem Definition

Let $\mathcal{D} = \{(x_i, y_i, M_i)\}_{i=1}^{N}$ be a dataset containing N test images, where $x_i \in \mathbb{R}^{n \times m}$ is the i-th image, y_i is its ground-truth label, and M_i is the corresponding ground-truth mask of the target object. Conventional evaluation methods primarily measure the classification accuracy of a model. Let $acc \in [0, 1]$ denote the classification accuracy of a model.

Two CNNs may achieve identical accuracy while relying on substantially different image regions. In particular, a model may achieve high accuracy while its Grad-CAM activation map reveals that the activated regions concentrated outside the target object. Hence, accuracy is insufficient to distinguish models according to their object-focused explanatory behavior.

We seek to quantify the alignment between a model's rationale and its predictions. Let H_i denote the Grad-CAM heatmap resized to the input image corresponding to image x_i and prediction $\hat{y}_i$. We define a scoring function $\text{LMOS}_i \in [0, 1]$ to measure the overlap ratio between the decision-relevant regions of H_i and the ground-truth object mask M_i. To indicate model trustworthiness and robustness, we address the following two sub problems in this paper.

Firstly, we want to evaluate CNN models considering both their accuracy (acc) and mean object-focused explanation quality (LMOS). Formally, the objective is to develop an evaluation measure T as a function of acc and LMOS: $T = \Psi(acc, \text{LMOS})$. We address this problem by the proposed Explainable Trustworthy Index (ETI).

Secondly, we want to incorporate CNNs behavioural changes in response to perturbations in model evaluation metric. Let $\mathcal{P} = \{p_1, p_2, \ldots, p_P\}$ denote a set of input perturbations. A robust CNN should not only preserve its predictive performance under any changes in $\mathcal{P}$ but also maintain a similar object-focused activation pattern simailar to the model's behavior on the corresponding perturbed image. Let S_{acc} and S_{LMOS} represent the overall accuracy stability (the model's stability to perform under purturbation) and LMOS stability (the model's stability to remain the object-focused for making predictions) of a model. Our second problem is to obtain an evaluation measure R that jointly captures these two forms of stability across the evaluated perturbations: $R = \Phi(S_{\text{acc}}, S_{\text{LMOS}})$. We propose Explainable Robustness Index (ERI) to address this problem. Table 1 describes the notations used in the article.

Table 1
Notation used in problem formulation.

Notation	Description
$\mathcal{D}$	Dataset of N images with labels and target masks
N	Number of images in the dataset
x_i	i-th input image
y_i	Ground-truth class label of x_i
$\hat{y}_i$	Predicted class label for x_i
M_i	Ground-truth binary mask of the target object in x_i
H_i	Grad-CAM heatmap for x_i and $\hat{y}_i$
LMOS_i	Object-focused localization score for x_i
LMOS	Mean LMOS over the evaluated dataset
acc	Classification accuracy of a CNN
$\mathcal{P}$	Set of P input perturbations
S_{acc}	Overall accuracy stability under perturbations
S_{LMOS}	Overall LMOS stability under perturbations
T	Generic evaluation measure based on acc and LMOS
R	Generic robustness measure based on S_{acc} and S_{LMOS}
$\Psi(\cdot)$	Function combining acc and LMOS
$\Phi(\cdot)$	Function combining S_{acc} and S_{LMOS}

3. System Overview

Figure 2 illustrates the proposed evaluation framework. It assesses CNNs along three dimensions: classification performance, the alignment of model explanations with ground-truth object regions, and the stability of both under image perturbations. The pipeline comprises data preparation and model training, explanation-based assessment, and robustness assessment.

Data preparation and model training. Each dataset is divided into training, validation, and test subsets. For every test image, the target leaf or flower region is manually annotated to produce a binary ground-truth mask. Eight CNN architectures are trained on the training subset, with the validation subset used for model selection, and classification accuracy is computed on the test subset.

Explanation-based assessment (ETI). For each test image, Grad-CAM generates a heatmap for the predicted class. The heatmap is normalized and binarized at a threshold τ. The threshold selection process is described in Section 5.4. LMOS quantifies the proportion of salient Grad-CAM activation that falls within the object region. The mean LMOS over the test set is combined with accuracy to produce ETI, which separates models that achieve similar accuracy but differ in how strongly they focus on the target object.

Robustness assessment (ERI). Nine perturbations simulating real-world image acquisition conditions are applied to the test images. The perturbations include noise, blur, motion blur, brightness, contrast, hue, gamma adjustment, compression artifacts, and occlusion. Each perturbed test set is evaluated with the same trained models and Grad-CAM procedure. Accuracy stability measures how well classification performance is preserved relative to the clean test set, and LMOS stability measures how well explanation alignment is preserved. The two are combined into an ERI

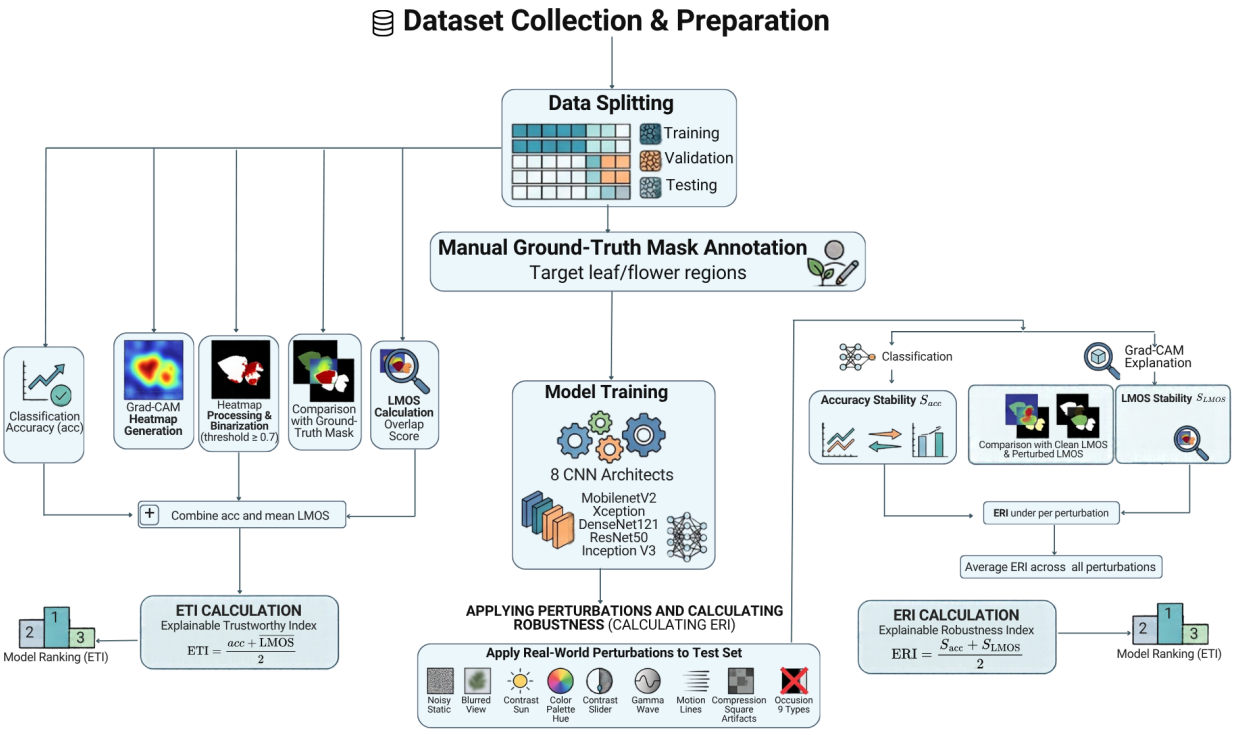


Figure 2: Overview of the ETI and ERI computation pipeline. Trained CNNs are evaluated on clean test images to obtain classification accuracy and Grad-CAM heatmaps. Thresholded heatmaps are compared with ground-truth object masks to compute LMOS, which is combined with accuracy to give ETI. Repeating the evaluation on perturbed test images yields accuracy stability and LMOS stability, which are combined to give ERI.

score for each perturbation, and these scores are averaged to give an overall ERI.

Models are ranked separately by ETI and by ERI. Because ETI characterizes behavior under normal conditions and ERI characterizes how much of that behavior survives perturbation, the two rankings need not agree (Section 6).

4. Explainable Trustworthy Index and Explainable Robustness Index

Both of the proposed Explainable Trustworthy Index (ETI) and Explainable Robustness Index (ERI) metrics use Localization Mask Overlap Score (LMOS) that measures the consistency between model-generated explanations and ground-truth object regions. ETI assesses if predictions are grounded in actual object regions rather than background noise. ERI extends this framework to perturbed inputs, measuring joint predictive and localization stability.

4.1. Localization Mask Overlap Score (LMOS)

LMOS calculates how much a model focuses on true object regions (instead of backgrounds) for prediction. The Grad-CAM heatmaps are generated from the last convolutional layer of a network. They are resized to the input image size. Let $H \in \mathbb{R}^{h \times w}$ denote the resized heatmap. We obtain a binary attention map $\hat{H}(i,j)$ from H by thresholding:

$$\hat{H}(i,j) = \mathbb{1}\big(H(i,j) > \tau\big) \tag{1}$$

where $\mathbb{1}(\cdot)$ is an indicator function and τ is the threshold. We set $\tau = 0.7$ (see Section 5.4) to restrict our LMOS calculations to red and yellow pixels, as the red and yelow pixels exhibit the highest activation levels in model predictions. Hence, $\hat{H}(i,j)$ is a binary mask representing the activated decision-relevant regions for the corresponding predictions and input image.

We define LMOS as the ratio of activated decision-relevant regions inside the object region and the total activated decision-relevant regions. Mathematically:

$$\text{LMOS} = \frac{\sum_{i=1}^{h}\sum_{j=1}^{w} \hat{H}(i,j)\, M(i,j)}{\sum_{i=1}^{h}\sum_{j=1}^{w} \hat{H}(i,j)} \tag{2}$$

where $M \in \{0,1\}^{h \times w}$ denote the ground-truth binary segmentation mask.

4.2. Explainable Trustworthy Index (ETI)

We propose Explainable Trustworthy Index (ETI) to measure trustworthiness of a CNN model's predictions. We combine the accuracy, explainable consistency and the mean LMOS (Eq. 2) to define ETI. Note that the LMOS is an instance specific (or local) measure. On the other hand accuray is a global measure indicating the overall performance of a model.

Let acc denote the classification accuracy of a model on test set and $\overline{\text{LMOS}}$ denote the mean LMOS scores across the same test samples. We define ETI as follows:

$$\text{ETI} = \frac{acc + \overline{\text{LMOS}}}{2}. \tag{3}$$

In this study, both acc and LMOS are equally important and are normalized to the range $[0, 1]$. Here, acc measures the overall predictive performance of a model, whereas LMOS measures the mean explanation consistency specific to the target object. We assign equal weighting to both components to avoid favoring either predictive performance or explanatory quality.

4.3. Explainable Robustness Index (ERI)

We estimate the robustness of a model by measuring its' performance deviation under purturbation. Let acc_c be the accuracy of a model on a clean dataset (not purturbed) and acc_p denote the accuracy obtained on the same dataset after applying a specific perturbation p (see Sec. 5.2 for all the purturbation methods used in this research). We intorduce the Accuracy Stability (S_{acc}) score under a specific perturbation p as:

$$S^p_{\text{acc}} = 1 - \frac{|acc_c - acc_p|}{acc_c + acc_p + \epsilon}$$

where ϵ is a small positive constant introduced to avoid division by zero error. A higher S^p_{acc} value indicates greater performance consistency under the perturbed condition p.

To estimate an overall measure of accuracy stability (S_{acc}), we compute S^p_{acc} across all $p \in \mathcal{P}$ where $\mathcal{P}$ is a set of purturbation methods, see Section 5.2, and take the mean of all S^p_{acc}. We define this as follows:

$$S_{\text{acc}} = \frac{1}{|\mathcal{P}|} \sum_{p=1}^{|\mathcal{P}|} S^p_{\text{acc}}$$

Similarly, we measure LMOS stability (S_{LMOS}) to quantify how consistently a model maintains its visual object focus under input perturbations relative to unperturbed images. For each sample in a dataset, we comput LMOS for both the original image and its perturbed version using Eq. 2.

Let LMOS^i_c be the LMOS value computed from sample i and LMOS^i_p be the LMOS value computed from the same sample i after applying perturbation p. To quantify the consistency between these two values, we define the LMOS stability under a perturbation p for sample i as:

$$S^{p,i}_{\text{LMOS}} = 1 - \frac{|\text{LMOS}^i_c - \text{LMOS}^i_p|}{\text{LMOS}^i_c + \text{LMOS}^i_p + \epsilon}$$

where ϵ is a small constant to avoid division by zero error. A higher value of $S^{p,i}_{\text{LMOS}}$ indicates that the model's reliance on the focus object remains more consistent under the corresponding perturbation.

Given there are N test images in a dataset, the LMOS stability under the perturbation p across the entire dataset is estimated as:

$$S^p_{\text{LMOS}} = \frac{1}{N} \sum_{i=1}^{N} S^{p,i}_{\text{LMOS}}.$$

To obtain an overall measure of LMOS stability across all evaluated perturbations $\mathcal{P}$, the perturbation-specific LMOS stability scores (S^p_{LMOS}) are subsequently averaged as:

$$S_{\text{LMOS}} = \frac{1}{|\mathcal{P}|} \sum_{p=1}^{\mathcal{P}} S^p_{\text{LMOS}}.$$

To jointly evaluate the robustness of predictive accuracy and explainability under input perturbations, we propose the *Explainable Robustness Index* (ERI). While accuracy stability and LMOS stability independently assess predictive consistency and visual localization behavior, ERI provides a unified robustness measure by integrating both dimensions.

The ERI under perturbation p (ERI^p) is then defined by combining the accuracy stability S^p_{acc} and LMOS stability S^p_{LMOS} under perturbation p as follows:

$$\text{ERI}^p = \frac{S^p_{\text{acc}} + S^p_{\text{LMOS}}}{2}. \tag{4}$$

The overall ERI is defined as the mean of overall accuracy stability and overall consistency of model's reliance on object focus as follows:

$$\text{ERI} = \frac{S_{\text{acc}} + S_{\text{LMOS}}}{2}. \tag{5}$$

Here, S_{acc} represents the overall consistency of classification accuracy between clean and perturbed inputs, while S_{LMOS} represents the overall consistency of the model's reliance on the focus object under perturbations. ERI jointly captures two complementary dimensions of robustness: *predictive stability* and *explanatory stability*.

Since both constituent scores share a common upper and lower bound, the resulting ERI serves as a normalized metric of overall robustness. A higher ERI value signifies that a model concurrently preserves both its classification accuracy and object-centric visual focus across input perturbations. Conversely, a lower ERI reflects marked instability in either predictive performance, visual focus, or both.

5. Experimental Setup

5.1. Datasets

We have used four publicly available datasets in our empirical study of the proposed evaluation measures:

- **EPLI**: EPLI (introduced as EgyPLI in [26]) contains 3,588 JPG images of size 256 × 256 pixels. There are images of 8 species including apple, berry, fig, guava, orange, plum, persimmon, and tomato. The dataset is annotated for image-level classification [26].
- **BDML:** BDML(also known as BDMediLeaves [29]) is a dataset of 2048 original leaf images (augmented to 38,606) from 10 medicinal plant species. They are captured with an iPhone 13 under natural light at 3024 × 4032 resolution. This dataset is designed for machine learning-based species identification. We used the original dataset (not the augmented one) in our experiments [29].
- **FLRS**: FLRS (or Flowers [30]) is a large image dataset for flower classification containing 3,670 training images of five flower species including daisies, tulips, sunflowers, dandelions, and roses. It is available through the TensorFlow Datasets (TFDS) collection [30]. The original images have varying spatial dimensions and are resized to the required input resolution of each CNN model during preprocessing.
- **FR**: This kaggle [31] dataset is very similar to the FLRS dataset above consisting the same five species (daisies, dandelions, roses, sunflowers, and tulips). However, the kaggle version contains 4,317 JPG images of resulation of 320 × 240 pixels. Although the image classes are the same as FLRS, the image samples are different. This dataset is used in previous research [32, 33].

Dataset preparation. The BDML dataset is already splitted into training, validation and testing sets. We split the other three datasets into training, validation and test subsets with an approximate ratio of 80%, 10% and 10%. Table 2 presents the class-wise training, validation, and test splits.

To simplify explainability and localization analysis, the test images have been manually annotated to generate corresponding JSON files. These JSON annotations describe the polygonal boundaries of the target leaf in each image. The JSON files have been converted into binary masks. Here, each mask represents only the target leaf region and excludes the background and other objects in the scene.

Table 2
Class-wise distribution of the experimental datasets.

Dataset / Class	Total	Train	Val	Test
EPLI				
Apple	519	415	52	52
Berry	340	272	34	34
Fig	508	406	50	52
Guava	520	416	52	52
Orange	547	437	55	55
Plum	468	374	47	47
Persimmon	527	421	53	53
Tomato	159	127	16	16
Subtotal	**3,588**	**2,868**	**359**	**361**
BDML				
Azadirachta indica	205	144	41	20
Calotropis gigantea	233	169	43	21
Centella asiatica	201	141	40	20
Hibiscus rosa-sinensis	214	150	43	21
Justicia adhatoda	200	140	40	20
Kalanchoe pinnata	229	160	46	23
Mikania micrantha	201	140	40	21
Ocimum tenuiflorum	151	106	30	15
Phyllanthus emblica	200	140	40	20
Terminalia arjuna	214	150	43	21
Subtotal	**2048**	**1440**	**406**	**202**
FLRS				
Daisy	633	506	64	63
Dandelion	898	718	90	90
Roses	641	512	64	65
Sunflowers	699	560	70	69
Tulips	799	639	80	80
Subtotal	**3,670**	**2,935**	**368**	**367**
FR				
Daisy	764	611	76	77
Dandelion	1052	842	105	105
Roses	784	627	78	79
Sunflowers	733	586	73	74
Tulips	984	787	98	99
Subtotal	**4,317**	**3,453**	**430**	**434**

5.2. Perturbation methods

To evaluate the robustness of the proposed framework under distribution shifts and common image corruptions, we have applied nine image perturbation methods to the test set. These types of perturbations were used for robustness measurement in previous works [16, 34]. The perturbation parameters have been fixed across all experiments to ensure reproducibility and enable fair comparisons among different CNN architectures. The details of the perturbations are given below:

- **Gaussian Blur:** Applied using a 5 × 5 kernel to simulate the loss of fine spatial details caused by defocus or low-quality imaging conditions.
- **Gaussian Noise:** Additive Gaussian noise with zero mean and standard deviation 0.05 has been introduced to model sensor noise and random intensity fluctuations.
- **Brightness Adjustment:** Increased by adding a constant value of 0.2 to normalized pixel intensities, after which values have been clipped to the valid range [0, 1].
- **Hue Shift:** Hue values in the HSV color space have been shifted by 15 units while preserving the saturation and value channels, simulating color variations caused by illumination or camera characteristics.
- **Contrast Adjustment:** Modified using mean-centered linear scaling: $I' = (I - \mu_I) \times 0.5 + \mu_I$, where I

denotes the input image and μ_I represents the mean pixel intensity (computed per channel). The resulting values have been clipped to the range [0, 1].

- **Gamma Correction:** Nonlinear illumination changes have been simulated using gamma correction: $I' = I^{2.0}$, where pixel intensities have been raised to the power of $\gamma = 2.0$.
- **Motion Blur:** Simulated using a 7×7 horizontal averaging kernel to emulate camera or object motion during image acquisition.
- **JPEG Compression:** Artifacts have been introduced by encoding and decoding the images with a quality factor of 30, simulating degradation due to lossy storage or transmission.
- **Random Occlusion:** A randomly positioned rectangular region corresponding to 20% of the image height and 20% of the image width has been masked by setting its pixel values to zero, thereby simulating partial object occlusion.

5.3. Implementaion the CNN models

In our experiments, we have used the following CNN models:

- **MobileNetV2**: MobileNetV2 is a lightweight CNN architecture designed to achieve a favorable trade-off between classification performance and computational efficiency. It employs inverted residual blocks with linear bottlenecks and lightweight depthwise convolutions to reduce computational cost while maintaining effective feature representations [35].
- **Xception**: Xception is based on depthwise separable convolutions and can be viewed as an extension of the Inception design in which spatial and channel-wise feature learning are more explicitly separated. The architecture provides an efficient use of model parameters while maintaining strong image classification performance [25].
- **DenseNet121**: DenseNet121 introduces dense connectivity, in which each layer receives feature maps from all preceding layers within a dense block. This design encourages feature reuse, strengthens feature propagation, and helps alleviate the vanishing-gradient problem while maintaining relatively efficient parameter usage [36].
- **InceptionResNetV2**: InceptionResNetV2 combines the multi-branch Inception architecture with residual connections to facilitate the training of deep networks. The residual connections accelerate optimization and enable effective learning in a computationally efficient Inception-based architecture [37].
- **VGG16**: VGG16 is a deep convolutional architecture that uses a simple and uniform design based primarily on stacked 3×3 convolutional filters followed by pooling layers. Its increased depth and straightforward architecture have made it a widely used baseline for image classification and transfer learning [38]
- **InceptionV3**: InceptionV3 employs factorized convolutions and carefully designed Inception modules to increase representational capacity while controlling computational cost. The architecture also incorporates auxiliary classification and other design improvements to achieve effective image recognition performance [39].
- **ResNet50**: ResNet50 employs residual learning through shortcut connections that allow layers to learn residual mappings with respect to their inputs. This design facilitates the optimization of deeper networks and enables the effective learning of hierarchical visual representations [40].
- **Custom CNN**: In this research, it has been developed as the original EPLI research [26].

5.4. Implementaion of ETI and ERI

We apply Grad-CAM [7] to obtain the heatmap H for output of each model (described above in Section 5.3). We resized the heatmaps to the corresponding model's input image size, e.g., for Xception it is 299×299 and for other models, it is 224×224.

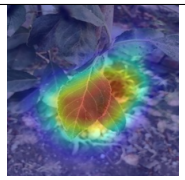
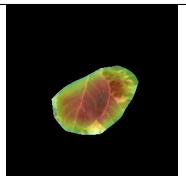
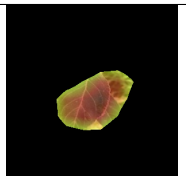
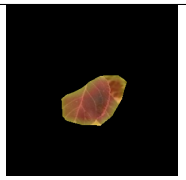
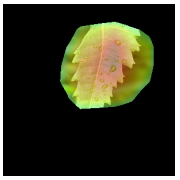
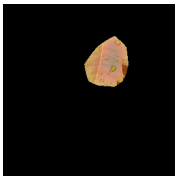
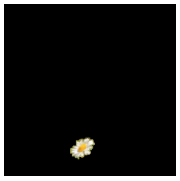
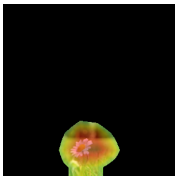
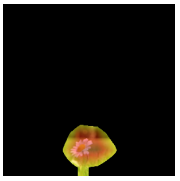
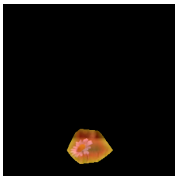
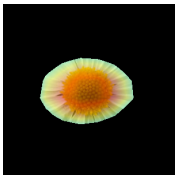
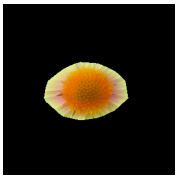
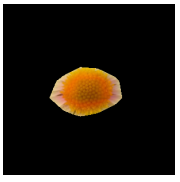
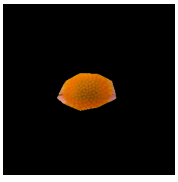
The next step in our method is to get the binary attention map $\hat{H}(i, j)$ for each sample image (see Eq. 1). We select the value of threshold τ through experiments are discussed below.

Using the binarized heatmap $\hat{H}(i, j)$ and the ground-truth binary mask $M(i, j)$, we compute LMOS according to Eq. 2. We created the ground-truth binary masks from the corresponding JSON files of test images and resized into the input image size (similar to the heatmaps). This same LMOS computation procedure is subsequently used to derive the proposed ETI and ERI scores, as defined in Eq. 3 and Eq. 5, respectively.

Selection of the threshold τ. The choice of the threshold parameter τ is critical, as it governs the filtering of decision-relevant regions during heatmap binarization. We experimented with $\tau \in \{0.5, 0.6, 0.7, 0.8\}$. Results of the experiments are shown in Table 3 including the original sample in the dataset (Original Image), the ground truth masks (Masked Image), Grad-CAM visualization (Grad-CAM Vis.) and effects of different τ as Threshold 0.5, Threshold 0.6, Threshold 0.7 and Threshold 0.8.

At $\tau = 0.5$, the binarized mask retains a broad portion of the visual attribution map capturing strong activity in red and yellow regions. It also illustrates substantial weak activity in the green regions. While this lower threshold preserves more total activations, incorporating these peripheral regions risks introducing irrelevant visual evidence into the masked Grad-CAM. Consequently, this raises concerns regarding whether

Table 3
Comparison of Grad-CAM masked visualizations under different threshold values.

Original Image	Masked Image	Grad-CAM Vis.	Threshold 0.5	Threshold 0.6	Threshold 0.7	Threshold 0.8
						
						
						
						

the resulting binary mask sufficiently isolates true decision-relevant regions.

Increasing $\tau = 0.6$ improves selectivity by suppressing most green activations, yet it permits weakly activated peripheral regions to pass alongside core object features. Conversely, $\tau = 0.8$ imposes overly stringent constraints, retaining predominantly red regions while discarding valid, decision-relevant yellow activations.

As illustrated in Table 3, $\tau = 0.7$ achieves an optimal balance by isolating the primary red and yellow heatmaps (representing the model's core decision-relevant evidence) while successfully suppressing low-intensity background noise. Based on this experiment, we select $\tau = 0.7$ for $\hat{H}(i, j)$ computation and applied consistently across all datasets, architectures, and perturbations to ensure cross-experimental comparability.

Variations of ETI. To demonstrate the impact of LMOS and *acc* on the proposed ETI, we have calculated two variants of ETI as folllows:

$$\mathrm{ETI}_{\text{w/o LMOS}} = \frac{\mathrm{acc}}{2}, \quad \mathrm{ETI}_{\text{w/o acc}} = \frac{\overline{\mathrm{LMOS}}}{2}.$$

Variations of ERI. To analyze the contribution of the individual components of ERI (i.e., S_{acc} and S_{LMOS}), we calculate two variants by removing one component at a time:

$$\mathrm{ERI}_{\text{w/o LMOS}} = \frac{S_{\mathrm{acc}}}{2}, \quad \mathrm{ERI}_{\text{w/o acc}} = \frac{S_{\mathrm{LMOS}}}{2}.$$

5.5. Evaluation metrics

We evaluate the CNN architectures using classification accuracy (acc), mean localization metric based on object focus (LMOS), the proposed Explainable Trustworthy Index (ETI), and the Explainable Robustness Index (ERI). Accuracy measures classification performance, while mean LMOS evaluates the alignment between Grad-CAM activations and ground-truth object masks. ETI combines accuracy and mean LMOS to assess classification performance and object-focus alignment, whereas ERI combines the stability of these two components under image perturbations. The definitions and computation procedures for LMOS, ETI, and ERI are provided in Section 5.4

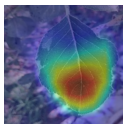
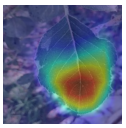
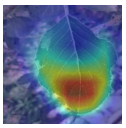
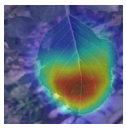
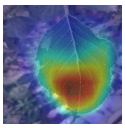
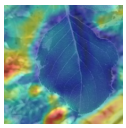
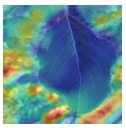
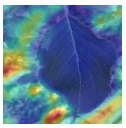
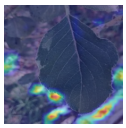
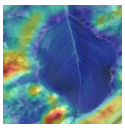
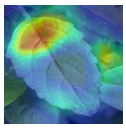
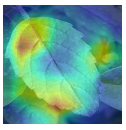
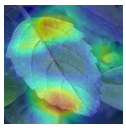
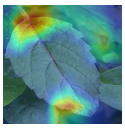
For comparative analysis, the CNN architectures are ranked in descending order of their ETI scores separately for each dataset, with Rank 1 assigned to the architecture achieving the highest ETI. This ranking considers both classification accuracy and object-focus alignment rather than classification accuracy alone. The ETI ablation variants are also reported to examine the contributions of accuracy and LMOS to ETI. Similarly, the ERI ablation variants are used to analyze the individual contributions of accuracy stability and LMOS stability to ERI.

5.6. Training configurations

We used TensorFlow/Keras on the Kaggle[1] environment with two NVIDIA Tesla T4 GPUs to trainining and evaluation of all the CNN models in our experiments. The

[1] kaggle.com/

Table 4
Grad-CAM visualization on EPLI and BDML datasets under clean and perturbed conditions.

Model Metrics	Original	True Mask (M)	Clean	Hue	Blur	Contrast	Compression
Model: Xception **Accuracy:** 99.45% **Mean LMOS:** 0.7638 **ETI:** 0.8791 **ERI:** 0.9346	 Apple (EPLI)		 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00
Model: Custom CNN **Accuracy:** 99.45% **Mean LMOS:** 0.1900 **ETI:** 0.5922 **ERI:** 0.7944	 Apple (EPLI)		 LMOS: 0.00	 LMOS: 0.00	 LMOS: 0.00	 LMOS: 0.00	 LMOS: 0.00
Model: Xception **Accuracy:** 95.05% **Mean LMOS:** 0.7800 **ETI:** 0.8653 **ERI:** 0.8982	 O. tenuiflorum (BDML)		 LMOS: 1.00	 LMOS: 1.00	 LMOS: 0.903	 LMOS: 0.812	 LMOS: 0.785
Model: Inception V3 **Accuracy:** 93.07% **Mean LMOS:** 0.3344 **ETI:** 0.6325 **ERI:** 0.7898	 O. tenuiflorum (BDML)		 LMOS: 0.383	 LMOS: 0.659	 LMOS: 0.477	 LMOS: 0.390*	 LMOS: 0.176

training configuration has followed the settings adopted in the original EPLI study [26]. The custom CNN was trained from scratch, while other models have been initialized with ImageNet-pretrained weights [41] and their backbone layers have been frozen.

For the pretrained models, the original classification heads have been removed and replaced with a common classification head consisting of Global Average Pooling [42] a dropout layer [43] with dropout rate 0.3, a fully connected layer with 128 ReLU units [44], and the final softmax layer has been configured according to the number of classes in each dataset. The Adam optimiser [45] has been used with a learning rate of 5×10^{-5} and categorical cross-entropy loss [42].

6. Results and Discussion

6.1. Grad-CAM visualization

Tables 4 and 5 present representative Grad-CAM visualizations with corresponding LMOS values across the four benchmark datasets. For each selected sample, the visualization strip contains the original input image, its ground-truth segmentation mask, the segmented target obtained by removing the background, and Grad-CAM visualizations under clean and perturbed conditions. These qualitative examples complement the quantitative ETI and ERI results by illustrating how different CNN architectures focus on the target object and how their attention patterns change under input perturbations.

On the EPLI dataset, Xception achieves a high mean LMOS of 0.7638 and an ETI of 0.8791, indicating that its classification decisions are generally associated with regions corresponding to the target leaf. As shown in Table 4, the Grad-CAM maps remain well aligned with the target region across the clean, hue, blur, contrast, and compression conditions, with an LMOS of 1.00 for the selected sample. In contrast, the Custom CNN achieves the same classification accuracy of 99.45% but obtains a substantially lower mean LMOS of 0.1900 and an ETI of 0.5922. Its representative Grad-CAM maps show an LMOS of 0.00 across the displayed conditions, demonstrating that high classification accuracy alone does not necessarily imply meaningful object-focused attention.

A similar pattern is observed on BDML. Xception obtains a mean LMOS of 0.7800 and an ETI of 0.8653, while InceptionV3 obtains considerably lower values of 0.3344 and 0.6325, respectively. For the selected BDML sample, Xception maintains strong localization under the clean and hue conditions, while its attention becomes less concentrated under blur, contrast, and compression, with LMOS values of 0.903, 0.812, and 0.785, respectively. InceptionV3 exhibits substantially weaker localization, with LMOS values ranging from 0.176 to 0.659 across the displayed conditions. This qualitative difference is consistent with the corresponding ETI values and demonstrates the ability of LMOS to distinguish models that may otherwise exhibit relatively competitive classification performance.

The flower datasets further demonstrate that the relationship between classification accuracy and explanation quality is not uniform across datasets. On FLRS, Xception achieves an ETI of 0.7849 with a mean LMOS of 0.6869, whereas InceptionV3 obtains an ETI of 0.5989 and a mean LMOS of 0.3749. The selected FLRS sample shows that Xception

Table 5
Grad-CAM visualization on Flower datasets (FLRS and FR).

Model Metrics	Original	True Mask (M)	Clean	Hue	Blur	Contrast	Compression
Model: Xception **Accuracy:** 88.28% **Mean LMOS:** 0.6869 **ETI:** 0.7849 **ERI:** 0.9308	 Rose (FLRS)		 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00
Model: Inception V3 **Accuracy:** 82.29% **Mean LMOS:** 0.3749 **ETI:** 0.5989 **ERI:** 0.8593	 Rose (FLRS)		 LMOS: 0.305	 LMOS: 0.219	 LMOS: 0.181	 LMOS: 0.219	 LMOS: 0.214
Model: Xception **Accuracy:** 90.32% **Mean LMOS:** 0.6029 **ETI:** 0.7531 **ERI:** 0.9689	 Daisy (FR)		 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00	 LMOS: 1.00
Model: Inception V3 **Accuracy:** 88.25% **Mean LMOS:** 0.3063 **ETI:** 0.5944 **ERI:** 0.8827	 Daisy (FR)		 LMOS: 0.633	 LMOS: 0.583	 LMOS: 0.550	 LMOS: 0.388	 LMOS: 0.505

maintains an LMOS of 1.00 across all displayed perturbations, whereas the InceptionV3 maps exhibit substantially lower localization scores, ranging from 0.181 to 0.305 for several perturbations. On FR, Xception again demonstrates stronger explanation quality, achieving a mean LMOS of 0.6029 and an ETI of 0.7531. Its selected sample maintains an LMOS of 1.00 under all displayed conditions, while InceptionV3 produces lower and more variable localization scores, ranging from 0.388 to 0.633.

Overall, these visualizations reinforce the main motivation of ETI: classification accuracy alone is insufficient to characterize the reliability of a CNN when the model may base its prediction on regions unrelated to the target object. Models such as Xception demonstrate that strong classification performance can be accompanied by substantially better target-focused attention, whereas the Custom CNN and, in several cases, InceptionV3 exhibit a noticeable discrepancy between predictive performance and localization quality. The qualitative Grad-CAM results therefore provide an intuitive interpretation of the quantitative LMOS and ETI measurements and illustrate how the proposed evaluation framework can expose differences in model behavior that conventional accuracy metrics may overlook.

6.2. Evaluation of the ETI

Table 6 illustrates a detailed evaluation of the proposed Explainable Trustworthiness Index (ETI) across four benchmark datasets: EPLI, BDML, FLRS, and FR. ETI not only considers accuracy but also explanation quality. It integrates prediction accuracy and the mean Localization Mask Overlap Score (LMOS). Therefore, a model achieves a higher ETI only when it possesses both reliable classification capability and consistent focus on semantically meaningful image regions instead of the background. The results clearly indicate that accuracy alone is not enough to measure model trustworthiness. Several models have achieved competitive classification performance but fail to achieve higher ETI values due to poor localization quality.

For the EPLI dataset, MobileNetV2 and Xception have achieved the highest ETI values of 0.8855 and 0.8791, respectively, despite DenseNet121 obtaining the highest classification accuracy of 1.0000. Although DenseNet121 perfectly classifies all test samples, its lower LMOS value (0.6964) reduces its overall ETI to 0.8482. This observation highlights the importance of explanation consistency as well as predictive accuracy for trustworthiness. Similarly, the Custom CNN achieves a high accuracy of 0.9945, which is comparable to the best-performing architectures. However, its LMOS value is the lowest (0.1900). So, ultimately, the ETI value is the lowest (0.5922). This indicates that the model may rely on irrelevant image regions rather than the focus leaf regions for decision-making.

A similar trend is observed on the BDML dataset. MobileNetV2 and Xception provide the highest ETI scores of 0.8657 and 0.8653, respectively, showing balanced classification capability and explanation reliability. In contrast, DenseNet121 achieves the highest accuracy among the evaluated models (0.9802), but its lower LMOS value (0.7118) limits its ETI performance (0.8460). The results further demonstrate that models with slightly lower accuracy can outperform highly accurate models when they generate more meaningful visual explanations. For example, Xception

Table 6
Comparison of classification accuracy, mean LMOS, ETI, and ablation variants across different CNN architectures on four benchmark datasets.

Dataset	Rank	Model	Acc	$\overline{\text{LMOS}}$	ETI	ETI w/o Acc.	ETI w/o LMOS
EPLI	1	MobileNetV2	0.9945	0.7765	0.8855	0.3882	0.4972
	2	Xception	0.9945	0.7638	0.8791	0.3819	0.4972
	3	DenseNet121	1.0000	0.6964	0.8482	0.3482	0.5000
	4	InceptionResNetV2	0.9889	0.5529	0.7709	0.2764	0.4945
	5	VGG16	0.9834	0.5268	0.7551	0.2634	0.4917
	6	InceptionV3	0.9917	0.3617	0.6767	0.1809	0.4958
	7	ResNet50	0.6122	0.7256	0.6689	0.3628	0.3061
	8	Custom CNN	0.9945	0.1900	0.5922	0.0950	0.4972
BDML	1	MobileNetV2	0.9703	0.7611	0.8657	0.3805	0.4851
	2	Xception	0.9505	0.7800	0.8653	0.3900	0.4752
	3	DenseNet121	0.9802	0.7118	0.8460	0.3559	0.4901
	4	VGG16	0.8960	0.5873	0.7416	0.2936	0.4480
	5	InceptionResNetV2	0.9059	0.5184	0.7122	0.2592	0.4530
	6	InceptionV3	0.9307	0.3344	0.6325	0.1672	0.4653
	7	ResNet50	0.4604	0.7817	0.6211	0.3909	0.2302
	8	Custom CNN	0.7376	0.3310	0.5343	0.1655	0.3688
FLRS	1	Xception	0.8828	0.6869	0.7849	0.3435	0.4414
	2	DenseNet121	0.9046	0.6188	0.7617	0.3094	0.4523
	3	MobileNetV2	0.8665	0.5958	0.7311	0.2979	0.4332
	4	VGG16	0.8147	0.6311	0.7229	0.3155	0.4074
	5	InceptionResNetV2	0.8501	0.5581	0.7041	0.2791	0.4251
	6	InceptionV3	0.8229	0.3749	0.5989	0.1875	0.4114
	7	ResNet50	0.5123	0.5623	0.5373	0.2812	0.2561
	8	Custom CNN	0.7003	0.2891	0.4947	0.1446	0.3501
FR	1	Xception	0.9032	0.6029	0.7531	0.3015	0.4516
	2	DenseNet121	0.9124	0.5382	0.7253	0.2691	0.4562
	3	VGG16	0.8387	0.5786	0.7087	0.2893	0.4194
	4	MobileNetV2	0.8986	0.5091	0.7038	0.2545	0.4493
	5	InceptionResNetV2	0.8894	0.4705	0.6800	0.2353	0.4447
	6	InceptionV3	0.8825	0.3063	0.5944	0.1531	0.4412
	7	ResNet50	0.4977	0.4769	0.4873	0.2384	0.2488
	8	Custom CNN	0.6820	0.2714	0.4767	0.1357	0.3410

achieves lower accuracy than DenseNet121 but obtains nearly the highest ETI due to superior localization behavior.

For the flower classification datasets, the ETI analysis reveals similar but dataset-dependent behavior. On the FLRS dataset, Xception achieves the highest ETI score of 0.7849, followed by DenseNet121 (0.7617) and MobileNetV2 (0.7311). Although DenseNet121 provides the highest accuracy (0.9046), Xception has achieved better overall ETI due to its higher LMOS value (0.6869). The FR also demonstrates the effectiveness of ETI in distinguishing models with similar predictive performance. Xception achieves the highest ETI score (0.7531), while DenseNet121 obtains the highest accuracy (0.9124). However, the lower LMOS of DenseNet121 (0.5382) reduces its ETI to 0.7253. These results confirm that the proposed metric provides additional insights beyond conventional accuracy ranking.

The ablation results shown in Table 6 further validate the contribution of both components of ETI. The "ETI without accuracy" variant, which considers only LMOS information, produces substantially lower values across all datasets because explanation quality alone cannot represent complete model reliability. Similarly, the "ETI without LMOS" variant, which depends only on classification accuracy, fails to capture whether the model focuses on meaningful regions.

Table 7
Accuracy stability of different CNN architectures across four datasets under various input perturbations.

Model	Noise	Blur	Brightness	Hue	Contrast	Gamma	Motion Blur	Compression	Occlusion	Mean
EPLI										
Custom CNN	0.9801	0.9958	0.9021	0.8711	0.8494	0.8033	0.9916	0.9986	0.9801	0.9302
DenseNet121	0.9944	0.9845	1.0000	0.9986	0.9860	0.9916	0.9159	0.9930	1.0000	0.9849
MobileNetV2	0.9441	0.9534	1.0000	0.9916	0.9902	0.9986	0.8816	0.9916	0.9830	0.9704
InceptionResNetV2	0.9727	0.9829	0.9986	0.9986	0.9944	0.9915	0.9149	0.9887	0.9469	0.9766
InceptionV3	0.9593	0.9771	0.9958	0.9972	0.9944	0.9915	0.9217	0.9757	0.9201	0.9703
ResNet50	0.9816	0.6962	0.9955	0.9862	0.4599	0.8978	0.5316	0.9769	0.9113	0.8263
VGG16	0.9828	0.9695	0.9958	0.9813	0.8380	0.9843	0.8284	0.9943	0.9481	0.9469
Xception	0.9801	0.9902	0.9986	1.0000	0.9972	0.9958	0.9549	0.9958	0.9859	0.9887
BDML										
Custom CNN	0.9689	0.7265	0.7319	0.8359	0.5174	0.8221	0.5943	0.9724	0.9084	0.7864
DenseNet121	0.9794	0.9496	0.9924	0.9846	0.9846	0.9898	0.8750	0.9714	0.9714	0.9665
MobileNetV2	0.9792	0.9111	0.9949	1.0000	0.9871	0.9949	0.8085	0.9574	0.9377	0.9523
InceptionResNetV2	0.9513	0.9360	0.9917	0.9917	0.9890	0.9973	0.8738	0.9204	0.9452	0.9552
InceptionV3	0.9810	0.9379	0.9838	0.9920	0.9865	0.9973	0.8876	0.9195	0.9038	0.9544
ResNet50	0.9780	0.4103	0.8929	0.9738	0.3966	0.8929	0.4103	0.9186	0.9249	0.7553
VGG16	1.0000	0.9538	0.9916	0.9803	0.8997	0.9745	0.8131	0.9888	0.9716	0.9526
Xception	0.9760	0.8739	0.9895	0.9921	0.9868	0.9948	0.8805	0.9122	0.9183	0.9471
FLRS										
Custom CNN	0.9902	0.9842	0.9553	0.8850	0.8527	0.9553	0.9741	0.9961	0.9822	0.9528
DenseNet121	0.9847	0.9510	0.9909	0.9657	0.9878	0.9785	0.9273	0.9816	0.9689	0.9707
MobileNetV2	0.9808	0.9539	0.9921	0.9921	0.9725	0.9905	0.9435	0.9742	0.9675	0.9741
InceptionResNetV2	0.9770	0.9703	0.9870	0.9887	0.9984	0.9935	0.9720	0.9804	0.9370	0.9783
InceptionV3	0.9967	0.9849	0.9950	0.9933	0.9917	0.9967	0.9477	0.9797	0.9883	0.9860
ResNet50	1.0000	0.8287	0.9556	0.9920	0.6714	0.9783	0.7871	0.9613	0.9838	0.9064
VGG16	0.9983	0.9725	0.9950	0.9672	0.9167	0.9966	0.9283	0.9967	0.9795	0.9723
Xception	0.9985	0.9843	0.9938	1.0000	0.9922	0.9985	0.9682	0.9763	0.9763	0.9876
FR										
Custom CNN	0.9933	0.9828	0.9504	0.8872	0.8851	0.9704	0.9811	0.9949	0.9614	0.9563
DenseNet121	0.9975	0.9767	0.9987	0.9781	0.9781	0.9898	0.9468	0.9741	0.9872	0.9808
MobileNetV2	0.9750	0.9402	0.9896	0.9883	0.9857	0.9935	0.9516	0.9791	0.9791	0.9758
InceptionResNetV2	0.9788	0.9802	0.9936	1.0000	0.9909	0.9909	0.9734	0.9869	0.9352	0.9811
InceptionV3	0.9841	0.9773	0.9961	0.9961	0.9841	0.9908	0.9521	0.9800	0.9550	0.9795
ResNet50	0.9640	0.8449	0.9714	0.9835	0.6584	0.9739	0.7514	0.9307	0.9714	0.8944
VGG16	0.9931	0.9555	0.9945	0.9761	0.9102	0.9832	0.8851	0.9986	0.9832	0.9644
Xception	0.9910	0.9962	0.9987	0.9975	0.9975	0.9897	0.9752	0.9831	0.9818	0.9901

For instance, in the EPLI dataset, the Custom CNN achieves an accuracy-based ETI value of 0.4972 when LMOS is removed, which incorrectly suggests competitive trustworthiness compared with stronger architectures. However, the complete ETI value decreases to 0.5922 due to its poor localization capability. This demonstrates that LMOS provides essential complementary information that cannot be obtained from accuracy alone.

Across all datasets, the ablation trends consistently show that the complete ETI formulation produces a more balanced and discriminative evaluation compared with either individual component. The integration of accuracy and LMOS prevents the possibility of overestimation of models that achieve high classification scores through potentially unreliable visual reasoning. Overall, the experimental results from Table 6 demonstrate that ETI provides a more comprehensive measurement of CNNs by simultaneously evaluating predictive reliability and explanation quality.

6.3. Evaluation of the ERI

The results in Tables 7 and 8 demonstrate that the stability of CNN models varies across datasets and perturbation types. Xception exhibits consistently high accuracy stability, achieving the highest mean values on EPLI (0.9887), FLRS (0.9876), and FR (0.9901). On BDML, DenseNet121 achieves the highest mean accuracy stability of 0.9665, followed by InceptionResNetV2 (0.9552) and InceptionV3 (0.9544). These results indicate that several ImageNet-pretrained architectures maintain relatively consistent classification performance under the evaluated image perturbations. However, accuracy stability alone does not provide a complete picture of model reliability, as a model may preserve its predicted class while its decision-relevant regions change substantially.

The mean LMOS stability results provide a complementary perspective. Xception achieves the highest mean LMOS stability on EPLI (0.8805), FLRS (0.8741), and FR (0.9478), whereas ResNet50 obtains the highest value on FR (0.8960). Notably, ResNet50 does not achieve the highest accuracy stability on any of the four datasets, although its LMOS stability is particularly strong on BDML and

Table 8
LMOS stability of different CNN architectures across four datasets under various input perturbations.

Model	Noise	Blur	Brightness	Hue	Contrast	Gamma	Motion Blur	Compression	Occlusion	Mean
EPLI										
Custom CNN	0.7525	0.7421	0.5343	0.5701	0.4545	0.5053	0.7242	0.9071	0.7376	0.6586
DenseNet121	0.8322	0.8151	0.9352	0.8723	0.8256	0.8398	0.6551	0.8103	0.8402	0.8251
MobileNetV2	0.8099	0.8200	0.9303	0.8910	0.8382	0.8763	0.7212	0.8393	0.8373	0.8404
InceptionResNetV2	0.6846	0.6978	0.8646	0.8556	0.7867	0.7693	0.5969	0.7433	0.7177	0.7463
InceptionV3	0.6550	0.6384	0.8425	0.8184	0.7378	0.7450	0.5560	0.6294	0.6426	0.6961
ResNet50	0.8417	0.7607	0.9332	0.9070	0.7830	0.8010	0.7666	0.8410	0.8187	0.8281
VGG16	0.7576	0.7014	0.8812	0.7836	0.5362	0.7432	0.5007	0.7693	0.6564	0.7033
Xception	0.8540	0.8611	0.9482	0.9489	0.9122	0.8802	0.7613	0.8712	0.8877	0.8805
BDML										
Custom CNN	0.6527	0.5391	0.5192	0.5206	0.4322	0.4802	0.5138	0.7388	0.6041	0.5556
DenseNet121	0.8505	0.7066	0.8788	0.8860	0.8255	0.8573	0.6538	0.8219	0.8365	0.8130
MobileNetV2	0.8044	0.7037	0.8849	0.8624	0.7967	0.8231	0.6132	0.7769	0.7908	0.7840
InceptionResNetV2	0.6655	0.6017	0.8148	0.7627	0.7656	0.7649	0.5920	0.6376	0.6674	0.6969
InceptionV3	0.6297	0.5752	0.7254	0.7291	0.6905	0.6274	0.5267	0.5578	0.5659	0.6253
ResNet50	0.9070	0.8051	0.8610	0.9044	0.7997	0.8418	0.8091	0.8880	0.8927	0.8565
VGG16	0.8378	0.6823	0.7996	0.7550	0.6200	0.7023	0.4370	0.7505	0.7047	0.6988
Xception	0.8693	0.7526	0.9185	0.9105	0.9097	0.8719	0.7333	0.8027	0.8747	0.8493
FLRS										
Custom CNN	0.7787	0.7326	0.6323	0.5899	0.5376	0.6056	0.6934	0.8218	0.6951	0.6763
DenseNet121	0.8577	0.7766	0.8921	0.8085	0.8384	0.8277	0.7809	0.8376	0.8382	0.8286
MobileNetV2	0.7818	0.7627	0.8826	0.8121	0.7751	0.8115	0.7459	0.7510	0.8251	0.7942
InceptionResNetV2	0.8068	0.7550	0.8578	0.8547	0.8234	0.8297	0.7444	0.7844	0.7600	0.8018
InceptionV3	0.7358	0.6701	0.8221	0.7992	0.7877	0.7356	0.6244	0.6879	0.7307	0.7326
ResNet50	0.9087	0.7728	0.8949	0.8864	0.7239	0.8407	0.7469	0.8621	0.8762	0.8347
VGG16	0.8757	0.8135	0.8687	0.8143	0.7450	0.8218	0.7380	0.8750	0.8403	0.8214
Xception	0.8573	0.8562	0.9176	0.9100	0.8993	0.8870	0.8249	0.8526	0.8619	0.8741
FR										
Custom CNN	0.8703	0.8340	0.7432	0.7218	0.6697	0.7233	0.8263	0.8870	0.7425	0.7798
DenseNet121	0.9124	0.8823	0.9376	0.8907	0.9138	0.9102	0.8383	0.9085	0.8867	0.8978
MobileNetV2	0.8420	0.8128	0.9088	0.8817	0.8623	0.8572	0.8231	0.8321	0.8666	0.8541
InceptionResNetV2	0.8762	0.8427	0.9283	0.8963	0.9031	0.8905	0.8281	0.8689	0.8281	0.8736
InceptionV3	0.7838	0.7453	0.8439	0.8416	0.8217	0.7972	0.7059	0.7697	0.7634	0.7858
ResNet50	0.9363	0.8639	0.9226	0.9181	0.8441	0.9022	0.8536	0.9011	0.9223	0.8960
VGG16	0.9321	0.8797	0.9242	0.8781	0.8303	0.9000	0.8449	0.9180	0.8846	0.8880
Xception	0.9425	0.9469	0.9591	0.9667	0.9607	0.9653	0.9154	0.9339	0.9398	0.9478

FLRS. Conversely, a model's high accuracy stability does not necessarily guarantee equally high localization stability. For example, InceptionV3 exhibits high mean accuracy stability on FLRS (0.9860), but its mean LMOS stability is comparatively lower (0.7326). This difference highlights the distinction between maintaining a classification decision and consistently focusing on the relevant image regions under perturbation. Therefore, examining both stability measures provide a more informative comparison of the evaluated models than considering either measure independently.

Table 9 provides a perturbation-specific comparison of the evaluated architectures across EPLI, BDML, FLRS, and FR. The results show that model performance depends not only on the dataset but also on the type of image perturbation. Although several architectures demonstrate strong performance across multiple perturbations, their reliability is not uniform under all forms of image degradation. Motion blur is particularly challenging in the reported experiments, while blur and contrast changes also expose weaknesses in some models, particularly on BDML. These findings suggest that the effect of a perturbation cannot be characterized solely by its impact on classification performance; changes in the localization of decision-relevant regions must also be considered. The perturbation-specific results consequently help identify cases in which a model maintains comparatively stable predictions but exhibits weaker explanation stability, as well as cases in which both aspects remain relatively stable.

The overall ERI rankings in Table 10 further reveal consistent differences among the eight CNN architectures. Xception achieves the highest overall ERI on all four datasets, with scores of 0.9346 on EPLI, 0.8982 on BDML, 0.9308 on FLRS, and 0.9689 on FR. Its performance across the four datasets demonstrates that it provides the strongest combined stability among the evaluated models under the experimental conditions. DenseNet121 is another consistently strong performer, ranking third on EPLI (0.9050) and second on BDML (0.8897), FLRS (0.8997), and FR (0.9393). MobileNetV2 ranks second on EPLI (0.9054) and third on BDML (0.8682), while ranking fifth on both FLRS (0.8842) and FR (0.9149). This variation indicates that its relative performance depends more strongly on the dataset than that of Xception and DenseNet121.

Table 9
ERI of different CNN architectures across four datasets under individual input perturbations.

Model	Noise	Blur	Brightness	Hue	Contrast	Gamma	Motion Blur	Compression	Occlusion	Overall ERI
EPLI										
Custom CNN	0.8663	0.8690	0.7182	0.7206	0.6520	0.6543	0.8579	0.9528	0.8589	0.7944
DenseNet121	0.9133	0.8998	0.9676	0.9355	0.9058	0.9157	0.7855	0.9017	0.9201	0.9050
MobileNetV2	0.8770	0.8867	0.9651	0.9413	0.9142	0.9375	0.8014	0.9155	0.9102	0.9054
InceptionResNetV2	0.8286	0.8403	0.9316	0.9271	0.8905	0.8804	0.7559	0.8660	0.8323	0.8614
InceptionV3	0.8072	0.8077	0.9192	0.9078	0.8661	0.8682	0.7389	0.8025	0.7813	0.8332
ResNet50	0.9117	0.7285	0.9644	0.9466	0.6215	0.8494	0.6491	0.9089	0.8650	0.8272
VGG16	0.8702	0.8355	0.9385	0.8824	0.6871	0.8638	0.6645	0.8818	0.8023	0.8251
Xception	0.9170	0.9256	0.9734	0.9745	0.9547	0.9380	0.8581	0.9335	0.9368	0.9346
BDML										
Custom CNN	0.8108	0.6328	0.6256	0.6783	0.4748	0.6512	0.5541	0.8556	0.7563	0.6710
DenseNet121	0.9150	0.8281	0.9356	0.9353	0.9051	0.9236	0.7644	0.8967	0.9040	0.8897
MobileNetV2	0.8918	0.8074	0.9399	0.9312	0.8919	0.9090	0.7109	0.8672	0.8643	0.8682
InceptionResNetV2	0.8084	0.7689	0.9033	0.8772	0.8773	0.8811	0.7329	0.7790	0.8063	0.8260
InceptionV3	0.8054	0.7566	0.8546	0.8606	0.8385	0.8124	0.7072	0.7387	0.7349	0.7898
ResNet50	0.9425	0.6077	0.8769	0.9391	0.5982	0.8674	0.6097	0.9033	0.9088	0.8059
VGG16	0.9189	0.8181	0.8956	0.8677	0.7599	0.8384	0.6251	0.8697	0.8382	0.8257
Xception	0.9227	0.8133	0.9540	0.9513	0.9483	0.9334	0.8069	0.8575	0.8965	0.8982
FLRS										
Custom CNN	0.8844	0.8584	0.7938	0.7374	0.6951	0.7805	0.8337	0.9089	0.8387	0.8146
DenseNet121	0.9212	0.8638	0.9415	0.8871	0.9131	0.9031	0.8541	0.9096	0.9035	0.8997
MobileNetV2	0.8813	0.8583	0.9374	0.9021	0.8738	0.9010	0.8447	0.8626	0.8963	0.8842
InceptionResNetV2	0.8919	0.8627	0.9224	0.9217	0.9109	0.9116	0.8582	0.8824	0.8485	0.8900
InceptionV3	0.8662	0.8275	0.9085	0.8962	0.8897	0.8661	0.7860	0.8338	0.8595	0.8593
ResNet50	0.9544	0.8008	0.9253	0.9392	0.6976	0.9095	0.7670	0.9117	0.9300	0.8706
VGG16	0.9370	0.8930	0.9319	0.8907	0.8308	0.9092	0.8332	0.9359	0.9099	0.8968
Xception	0.9279	0.9203	0.9557	0.9550	0.9457	0.9427	0.8965	0.9144	0.9191	0.9308
FR										
Custom CNN	0.9318	0.9084	0.8468	0.8045	0.7774	0.8469	0.9037	0.9409	0.8519	0.8680
DenseNet121	0.9549	0.9295	0.9682	0.9344	0.9459	0.9500	0.8925	0.9413	0.9369	0.9393
MobileNetV2	0.9085	0.8765	0.9492	0.9350	0.9240	0.9254	0.8874	0.9056	0.9228	0.9149
InceptionResNetV2	0.9275	0.9114	0.9609	0.9482	0.9470	0.9407	0.9007	0.9279	0.8817	0.9273
InceptionV3	0.8840	0.8613	0.9200	0.9188	0.9029	0.8940	0.8290	0.8749	0.8592	0.8827
ResNet50	0.9502	0.8544	0.9470	0.9508	0.7512	0.9381	0.8025	0.9159	0.9468	0.8952
VGG16	0.9626	0.9176	0.9594	0.9271	0.8702	0.9416	0.8650	0.9583	0.9339	0.9262
Xception	0.9667	0.9715	0.9789	0.9821	0.9791	0.9775	0.9453	0.9585	0.9608	0.9689

InceptionResNetV2 also demonstrates relatively consistent performance, ranking fourth on EPLI (0.8614) and BDML (0.8260), and fourth on FLRS (0.8900) and third on FR (0.9273). VGG16 exhibits greater variation in its relative ranking, placing seventh on EPLI (0.8251), fifth on BDML (0.8257), third on FLRS (0.8968), and fourth on FR (0.9262). ResNet50 and InceptionV3 show more moderate overall ERI scores across the datasets. In particular, InceptionV3 ranks fifth on EPLI (0.8332), seventh on BDML (0.7898), seventh on FLRS (0.8593), and seventh on FR (0.8827). These results illustrate that high classification stability does not necessarily translate into a similarly high overall ERI when localization stability is also taken into account.

In contrast, the Custom CNN achieves the lowest overall ERI on every dataset, with scores of 0.7944 on EPLI, 0.6710 on BDML, 0.8146 on FLRS, and 0.8680 on FR. Its particularly low score on BDML indicates a substantial weakness under the combined evaluation of classification and localization stability. Although the Custom CNN exhibits relatively high mean accuracy stability on some datasets, its comparatively lower LMOS stability limits its overall performance. This observation reinforces the importance of evaluating the stability of model explanations alongside classification predictions. Nevertheless, the rankings reflect the tested datasets, architectures, and perturbations rather than establishing a universal ordering of CNN models.

The ablation results in Table 10 provide further insight into the relative contributions of accuracy stability and LMOS stability. The results show that the two components can contribute differently to the overall scores across architectures and datasets. Models with strong accuracy stability may obtain lower scores when their localization stability is comparatively weak, whereas models with strong localization stability may be limited by less stable classification performance. The contrast between InceptionV3's high accuracy stability on FLRS and its lower LMOS stability illustrates this distinction. Similarly, ResNet50's leading mean LMOS stability on BDML does not result in the highest overall ERI on that dataset. The ablation results therefore help explain why the overall rankings do not simply follow the rankings obtained from either individual stability measure.

Table 10
ERI ranking and ablation analysis of CNN architectures across four datasets.

Rank	Model	S_{acc}	S_{LMOS}	ERI	$ERI_{w/o\ acc}$	$ERI_{w/o\ LMOS}$
EPLI						
1	Xception	0.9887	0.8805	0.9346	0.4403	0.4944
2	MobileNetV2	0.9704	0.8404	0.9054	0.4202	0.4852
3	DenseNet121	0.9849	0.8251	0.9050	0.4125	0.4924
4	InceptionResNetV2	0.9766	0.7463	0.8614	0.3731	0.4883
5	InceptionV3	0.9703	0.6961	0.8332	0.3481	0.4852
6	ResNet50	0.8263	0.8281	0.8272	0.4140	0.4132
7	VGG16	0.9469	0.7033	0.8251	0.3517	0.4735
8	Custom CNN	0.9302	0.6586	0.7944	0.3293	0.4651
BDML						
1	Xception	0.9471	0.8493	0.8982	0.4246	0.4736
2	DenseNet121	0.9665	0.8130	0.8897	0.4065	0.4832
3	MobileNetV2	0.9523	0.7840	0.8682	0.3920	0.4762
4	InceptionResNetV2	0.9552	0.6969	0.8260	0.3485	0.4776
5	VGG16	0.9526	0.6988	0.8257	0.3494	0.4763
6	ResNet50	0.7553	0.8565	0.8059	0.4283	0.3777
7	InceptionV3	0.9544	0.6253	0.7898	0.3127	0.4772
8	Custom CNN	0.7864	0.5556	0.6710	0.2778	0.3932
FLRS						
1	Xception	0.9876	0.8741	0.9308	0.4371	0.4938
2	DenseNet121	0.9707	0.8286	0.8997	0.4143	0.4854
3	VGG16	0.9723	0.8214	0.8968	0.4107	0.4862
4	InceptionResNetV2	0.9783	0.8018	0.8900	0.4009	0.4891
5	MobileNetV2	0.9741	0.7942	0.8842	0.3971	0.4871
6	ResNet50	0.9064	0.8347	0.8706	0.4174	0.4532
7	InceptionV3	0.9860	0.7326	0.8593	0.3663	0.4930
8	Custom CNN	0.9528	0.6763	0.8146	0.3382	0.4764
FR						
1	Xception	0.9901	0.9478	0.9689	0.4739	0.4950
2	DenseNet121	0.9808	0.8978	0.9393	0.4489	0.4904
3	InceptionResNetV2	0.9811	0.8736	0.9273	0.4368	0.4905
4	VGG16	0.9644	0.8880	0.9262	0.4440	0.4822
5	MobileNetV2	0.9758	0.8541	0.9149	0.4270	0.4879
6	ResNet50	0.8944	0.8960	0.8952	0.4480	0.4472
7	InceptionV3	0.9795	0.7858	0.8827	0.3929	0.4898
8	Custom CNN	0.9563	0.7798	0.8680	0.3899	0.4781

Overall, the results across the four datasets consistently identify Xception as the strongest model under the combined evaluation, with DenseNet121 also demonstrating robust and comparatively consistent performance. The weaker overall performance of the Custom CNN, together with the differences between accuracy stability and LMOS stability across individual architectures, demonstrates that classification consistency alone may overlook important changes in model explanations. The perturbation-specific and ablation results further emphasize the value of evaluating both aspects when investigating the reliability of CNN predictions under changes in input images.

6.4. Cross-dataset analysis

The cross-dataset results provide a broader assessment of the proposed ETI and ERI metrics and reveal several consistent patterns across different visual domains. As shown in Tables 6 and the corresponding ERI results, the relative performance of the CNN architectures is influenced not only by their classification capability but also by the quality and stability of their visual explanations. While ETI evaluates a model by jointly considering classification accuracy and mean LMOS, ERI focuses on the stability of both prediction and explanation under input perturbations. Therefore, the two metrics provide complementary perspectives on CNN reliability.

Across the four datasets, ETI shows a clear distinction between the two leaf datasets and the two flower datasets. The highest ETI values are observed on the leaf datasets, EPLI and BDML. Here, MobileNetV2 has achieved 0.8855 and 0.8657, respectively, while Xception has achieved 0.8791 and 0.8653. In comparison, the highest ETI values on the FLRS and FR are 0.7849 and 0.7531, respectively, both achieved by Xception. This difference indicates that the models generally achieve stronger combinations of classification accuracy and localization quality on the leaf datasets than on the flower datasets. The lower ETI values on the flower datasets are mainly associated with reduced LMOS values, indicating that accurately recognizing flower categories does not necessarily imply that the models consistently focus on the corresponding flower regions.

Xception demonstrates particularly strong cross-dataset consistency for ETI. It ranks second on EPLI and BDML

and first on both flower datasets. More importantly, Xception maintains relatively high LMOS values across all four datasets, ranging from 0.6029 on FR to 0.7800 on BDML. This consistency allows Xception to maintain a high ETI even when another architecture achieves slightly higher classification accuracy. For example, DenseNet121 has achieved the highest accuracy on EPLI, BDML, FLRS, and FR; especially on EPLI, it reaches 1.0000, yet it does not obtain the highest ETI on any of the four datasets. This provides strong cross-dataset evidence that classification accuracy alone is insufficient for assessing model trustworthiness. On EPLI, the Custom CNN has achieved 99.45 accuracy, but its object reliance is the lowest (only 0.19). ETI of the Custom CNN is always low, even when its accuracy is high.

The cross-dataset ERI results reveal a related but distinct pattern. Xception achieves the highest overall ERI on all four datasets, with values of 0.9346 on EPLI, 0.8982 on BDML, 0.9308 on FLRS and 0.9689 on FR. It's consistently strong performance in both accuracy stability and LMOS stability indicates that the architecture is comparatively resilient to input perturbations while preserving its explanation patterns. Particularly notable is its performance on the FR, where Xception achieves an accuracy stability of 0.9901 and an LMOS stability of 0.9478, resulting in an ERI of 0.9689. Furthermore, Xception remains the strongest model across individual perturbations on this dataset, demonstrating that its robustness is not attributable to a small number of favorable transformations.

An important observation from the cross-dataset comparison is that ETI and ERI do not necessarily produce identical performance patterns. The FLRS dataset, for example, has a substantially lower ETI for Xception (0.7849) than EPLI (0.8791), but its ERI is very high (0.9308). Similarly, the FR produces the highest ERI for Xception (0.9689) despite having a lower ETI (0.7531). This difference is expected because ETI incorporates the absolute classification accuracy and mean LMOS, whereas ERI evaluates how consistently these properties are preserved when the input is perturbed. Consequently, a model may have moderate absolute performance but exhibit highly stable predictions and explanations, resulting in a high ERI. Thus, ETI and ERI should be interpreted as complementary rather than interchangeable measures.

The results also demonstrate that weak explanatory behavior can remain a major limitation even when classification performance is high. The Custom CNN provides a clear example across the datasets. On EPLI, it achieves an accuracy of 0.9945 but has a very low LMOS of 0.1900, resulting in the lowest ETI of 0.5922. A similar pattern is observed on BDML, FLRS, and the FR, where the Custom CNN consistently obtains among the lowest LMOS and ETI values. Its ERI performance is also comparatively weak, particularly because its explanation stability remains lower than that of the strongest pretrained architectures. These findings suggest that the proposed metrics can identify models whose apparent predictive performance is not accompanied by reliable or stable visual reasoning.

ResNet50 provides another useful cross-dataset example of the complementary nature of the two metrics. It exhibits relatively high LMOS or LMOS stability in several datasets but suffers from substantially lower classification accuracy or accuracy stability. For instance, its accuracy on BDML, FLRS, and FR is only 0.4604, 0.5123, and 0.4977, respectively. Consequently, its ETI remains low despite comparatively reasonable localization scores. In the ERI evaluation, ResNet50 similarly demonstrates that strong explanation stability alone cannot compensate for poor predictive stability. This reinforces the central principle of the proposed framework: reliable CNN behavior requires both predictive and explanatory consistency.

Overall, the cross-dataset analysis demonstrates that ETI and ERI capture different but complementary aspects of CNN reliability. ETI is particularly effective for distinguishing models according to their joint classification performance and explanation quality, whereas ERI assesses whether both properties remain stable under realistic input perturbations. Xception consistently emerges as the most reliable architecture across the four datasets, achieving the highest ERI on every dataset and the highest ETI on both flower datasets. In contrast, models such as the Custom CNN can achieve high classification accuracy while exhibiting substantially weaker explanation quality and robustness. These consistent cross-dataset observations provide further evidence that evaluating CNNs through accuracy alone can overlook important aspects of model reliability, while the combined use of ETI and ERI offers a more comprehensive assessment of trustworthy and robust visual recognition.

Despite the strengths of this study, there are some limitations. ETI and ERI consider only two properties: accuracy and object focus alignment. To evaluate the whole scenario, many properties may need to be considered. Some research raises questions about Explainable AI methods such as Grad-CAM [46, 47, 48]. In the future, we have a plan to extend ETI and ERI to other XAI methods such as Grad-CAM++[47], Score-CAM[49], LayerCAM[50], and Rise [51] to assess the sensitivity of the proposed metrics to the choice of explainability technique. We will try to adapt ETI and ERI for other computer vision tasks, including object detection, semantic segmentation, and vision-language models.

7. Related Works

7.1. Model Reliability and Trustworthiness

To move beyond standard classification accuracy, recent studies have increasingly focused on quantifying model reliability through uncertainty estimation and calibration. Javed et al. [14] integrated Bayesian Monte Carlo Dropout with nonparametric Conformal Prediction to address overconfidence in deep neural networks. To better evaluate model performance under uncertainty, Ao et al. [15] introduced the Excess Area Under the Optimal Risk-Coverage Curve (E-AUoptRC) and the Trust Index (TI), offering a risk-coverage perspective on model trustworthiness.

At a broader level, Nastoska et al. [52] provided a comprehensive review of AI trustworthiness criteria by evaluating industrial compliance against the NIST AI Risk Management Framework and ISO/IEC standards. In domain-specific applications, Pal et al. [53] proposed the Trust-Aware Explainable Artificial Intelligence (TAXAI) framework, which yields a normalized score by combining algorithmic transparency, interpretability alignment, and compliance reliability. While these frameworks advance the quantification of predictive trust, they generally evaluate reliability without explicitly measuring the spatial fidelity or stability of visual explanations under environmental perturbations.

7.2. Predictive Robustness Under Perturbations

Evaluating model performance under distribution shifts and input corruptions is essential for reliable deployment. Hendrycks and Dietterich [16] established foundational benchmarks known as ImageNet-C and ImageNet-P, to measure classification degradation under common corruptions and perturbations. Extending this, Schneider et al. [54] demonstrated that test-time covariate-shift adaptation can substantially recover predictive performance under corruptions, while Hendrycks et al. [17] proposed AugMix to improve model robustness while preserving clean accuracy. Beyond common corruptions, Su et al. [19] and Taori et al. [18] investigated adversarial and natural distribution shifts, demonstrating that high standard accuracy does not guarantee robustness and advocating for independent robustness evaluations. Furthermore, Guo et al. [20] compiled a 23-metric framework targeting adversarial data and model vulnerabilities.

While these works establish rigorous protocols for evaluating accuracy under stress, they focus exclusively on output predictions. A few recent efforts have attempted to incorporate explainability into robustness routines. Zhang et al. [55] proposed AR^2, aligning Class Activation Maps (CAMs) between clean and corrupted images during training to enforce feature consistency. However, AR^2 operates as a model repair mechanism rather than a quantitative evaluation index. In contrast, our proposed Explainable Robustness Index (ERI) directly quantifies the joint stability of both predictive performance and Grad-CAM visual evidence under input perturbations.

7.3. Explanation Quality, Localization, and Spurious Bias

Visual explanation techniques e.g., Grad-CAM, Attention maps, and Layer-wise Relevance Propagation (LRP) are widely deployed to interpret CNN decisions in sensitive domains, including smart farming and plant pathology [56, 57, 58]. However, as highlighted by Shawon et al. [58], high accuracy and visual heatmaps do not automatically guarantee model generalization under real-world variability. A critical challenge remains the reliance on shortcut learning and background bias. Bassi et al. [11] demonstrated that synthetic background biases heavily influence deep models, showing that LRP heatmaps can expose and mitigate shortcut learning without incurring high computational costs. Similarly, Zheng et al. [59] introduced ShortcutProbe to detect and retrain networks against prediction shortcuts without manual annotations.

A fundamental bottleneck in Explainable AI (XAI) is the scarcity of quantitative metrics for evaluating explanation quality. Addressing this, Rahimiaghdam and Alemdar [22] formulated evaluative metrics for medical imaging, while Baniecki et al. [60] proposed aligning model explanation pipelines directly with ground-truth segmentation masks. Ground-truth-based evaluation was further advanced by Guidotti et al. [24], who evaluated local explanation methods against annotated object bounds. Inspired by this perspective, our proposed Localization Mask Overlap Score (LMOS) quantifies the spatial alignment between Grad-CAM visual evidence and target object regions.

Regarding explanation stability, Repetto et al. [21] systematically measured the sensitivity of XAI techniques under natural noise and adversarial manipulations that alter explanations without affecting predictions. Finally, Fong and Vedaldi [23] and Kim et al. [61] analyzed black-box attribution and weakly supervised localization, observing that CAM-based methods often concentrate on localized, highly discriminative parts rather than full object extents. Our framework builds upon these insights by offering a unified, quantitative evaluation of spatial overlap (LMOS), explanation quality (ETI), and perturbation stability (ERI).

8. Conclusion

This paper has proposed two novel metrics, the Explainable Trustworthy Index (ETI) and the Explainable Robustness Index (ERI), to evaluate CNN models beyond classification accuracy. ETI combines accuracy with the Localization Mask Overlap Score (LMOS) to assess whether a model's predictions are grounded in the intended object rather than background or irrelevant regions. ERI extends this evaluation to perturbed inputs by jointly measuring accuracy stability and LMOS stability, capturing whether both predictive performance and object-focused reasoning remain consistent under real-world degradations.

Across four benchmark datasets and eight CNN architectures, our results demonstrate that high classification accuracy does not guarantee high ETI or ERI. For example, on the EPLI dataset, the Custom CNN model has achieved competitive or near-perfect accuracy while obtaining the lowest ETI and ERI scores due to poor and unstable object localization and its poor stability Similarly, the InceptionV3 achieves high classification accuracy across most conditions, its low ETI score reflects poor spatial focus on target object regions. Conversely, Xception consistently has achieved the highest overall ERI across all four datasets and the highest ETI on both flower datasets (FIRS and FR), despite not always achieving the highest raw accuracy. It illustrates that predictive performance and explanation quality/stability are related but distinct properties of model reliability. MobileNetV2 has outperformed higher-accuracy models on the BDML dataset in terms of ETI, reinforcing this distinction.

The cross-dataset comparison further shows that reliability patterns vary with visual domain: leaf datasets (EPLI and BDML) have generally yielded higher ETI than flower datasets (FLRS and FR), primarily due to lower LMOS on flowers, yet this does not translate into lower robustness. Several models have attained high ERI on the flower datasets despite modest ETI. This confirms that clean-image trustworthiness (ETI) and perturbation robustness (ERI) are complementary rather than interchangeable measures of reliability.

Overall, ETI and ERI provide a more comprehensive framework for evaluating CNNs in safety-critical or trust-sensitive applications, where accuracy alone is an insufficient proxy for whether a model reasons about the correct visual evidence and whether that reasoning holds up under realistic input perturbations.

Data and Code availability

Supplementary code and data (including `.json` files and `.keras` files) associated with this research are available at:
`https://github.com/ragib1996/ETI_ERI`

CRediT authorship contribution statement

Ragib Mehedi: Conceptualization, Methodology, Software, Investigation, Data Curation, Formal analysis, Writing – Original Draft, Writing – Review & Editing. **Azmain Inquaid Haque:** Investigation, Data Curation, Software, Validation. **Md. Nazmul Abdal:** Data Curation. **Abdullah Al Noman:** Supervision, Writing – Review & Editing. **Md. Zahidul Islam:** Supervision, Writing – Review & Editing .

Acknowledgments

The authors are always grateful to the Computer Science and Engineering Discipline, Khulna University, for their support and research facilities.

Disclosure statement

No conflict of interest was reported by the authors.

Funding

This research received no external funding.

References

[1] R. Archana and P. E. Jeevaraj, "Deep learning models for digital image processing: a review," *Artificial intelligence review*, vol. 57, no. 1, p. 11, 2024.

[2] D. Shen, G. Wu, and H.-I. Suk, "Deep learning in medical image analysis," *Annual review of biomedical engineering*, vol. 19, pp. 221–248, 2017.

[3] A. Abade, P. A. Ferreira, and F. de Barros Vidal, "Plant diseases recognition on images using convolutional neural networks: A systematic review," *Computers and Electronics in Agriculture*, vol. 185, p. 106125, 2021.

[4] F. Lanusse *et al.*, "The dawes review 10: The impact of deep learning for the analysis of galaxy surveys," *Publications of the Astronomical Society of Australia*, vol. 40, p. e001, 2023.

[5] A. Zafar, L. K. Murali, and A. Vashist, "Beyond accuracy: Evaluating visual grounding in multimodal medical reasoning," *arXiv preprint arXiv:2603.03437*, 2026.

[6] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," *arXiv preprint arXiv:1312.6199*, 2013.

[7] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE international conference on computer vision*, pp. 618–626, 2017.

[8] J. Wu, W. Kang, H. Tang, Y. Hong, and Y. Yan, "On the faithfulness of vision transformer explanations," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10936–10945, IEEE, 2024.

[9] A. Nguyen, J. Yosinski, and J. Clune, "Deep neural networks are easily fooled: High confidence predictions for unrecognizable images," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 427–436, 2015.

[10] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel, "Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness," in *International conference on learning representations*, 2018.

[11] P. R. Bassi, S. S. Dertkigil, and A. Cavalli, "Improving deep neural network generalization and robustness to background bias via layer-wise relevance propagation optimization," *Nature Communications*, vol. 15, no. 1, p. 291, 2024.

[12] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[13] K. Xiao, L. Engstrom, A. Ilyas, and A. Madry, "Noise or signal: The role of image backgrounds in object recognition," *arXiv preprint arXiv:2006.09994*, 2020.

[14] F. Javed, S. Ruijs, and A. Kosiakova, "Beyond accuracy: Reliability and uncertainty estimation in convolutional neural networks," *Available at SSRN 6030753*.

[15] S. Ao, S. Rueger, and A. Siddharthan, "Empirical optimal risk to quantify model trustworthiness for failure detection," *arXiv preprint arXiv:2308.03179*, 2023.

[16] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," *arXiv preprint arXiv:1903.12261*, 2019.

[17] D. Hendrycks, N. Mu, E. D. Cubuk, B. Zoph, J. Gilmer, and B. Lakshminarayanan, "Augmix: A simple data processing method to improve robustness and uncertainty," *arXiv preprint arXiv:1912.02781*, 2019.

[18] R. Taori, A. Dave, V. Shankar, N. Carlini, B. Recht, and L. Schmidt, "Measuring robustness to natural distribution shifts in image classification," *Advances in neural information processing systems*, vol. 33, pp. 18583–18599, 2020.

[19] D. Su, H. Zhang, H. Chen, J. Yi, P.-Y. Chen, and Y. Gao, "Is robustness the cost of accuracy?–a comprehensive study on the robustness of 18 deep image classification models," in *European Conference on Computer Vision*, pp. 644–661, Springer, 2018.

[20] J. Guo, W. Bao, J. Wang, Y. Ma, X. Gao, G. Xiao, A. Liu, J. Dong, X. Liu, and W. Wu, "A comprehensive evaluation framework for deep model robustness," *Pattern Recognition*, vol. 137, p. 109308, 2023.

[21] S. Repetto, I. Maljkovic, M. Lotto, A. E. Cinà, S. Vascon, and F. Roli, "Evaluating the robustness of explainable ai in medical image recognition under natural and adversarial data corruption," *Machine Learning*, vol. 115, no. 1, p. 4, 2026.

[22] S. Rahimiaghdam and H. Alemdar, "Evaluating the quality of visual explanations on chest x-ray images for thorax diseases classification," *Neural Computing and Applications*, vol. 36, no. 17, pp. 10239–10255, 2024.

[23] R. C. Fong and A. Vedaldi, "Interpretable explanations of black boxes by meaningful perturbation," in *2017 IEEE international conference on computer vision (ICCV)*, pp. 3449–3457, IEEE, 2017.

[24] R. Guidotti, "Evaluating local explanation methods on ground truth," *Artificial Intelligence*, vol. 291, p. 103428, 2021.

[25] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *2017 IEEE conference on computer vision and pattern recognition (CVPR)*, pp. 1800–1807, Ieee, 2017.

[26] A. M. Sarhan and M. A. Shaheen, "Egypli: A real-life annotated image dataset for egyptian plant leaf identification," *Scientific Data*, 2026.

[27] I. Kim, S. Rajaraman, and S. Antani, "Visual interpretation of convolutional neural network predictions in classifying medical image modalities," *Diagnostics*, vol. 9, no. 2, p. 38, 2019.

[28] A. S. M. Sayeed, S. A. Birahim, M. S. Ullah, Y. Islam, A. Paul, M. A. Hasan, M. Ahsan, and M. Kowalski, "Explainable and secure federated learning for privacy-enhancing skin cancer classification using a lightweight multi-scale cnn," *Scientific Reports*, vol. 16, no. 1, p. 11414, 2026.

[29] S. Islam, M. R. Ahmed, S. Islam, M. M. A. Rishad, S. Ahmed, T. R. Utshow, and M. I. Siam, "Bdmedileaves: A leaf images dataset for bangladeshi medicinal plants identification," *Data in Brief*, vol. 50, p. 109488, 2023.

[30] T. T. Team, "Flowers." `http://download.tensorflow.org/example_images/flower_photos.tgz`, jan 2019.

[31] A. Mamaev, "Flowers recognition | kaggle," 2018. https://www.kaggle.com/datasets/alxmamaev/flowers-recognition (accessed July 13, 2026).

[32] S. Balasubaramanian, M. A. Shah, *et al.*, "Restricted boltzmann machine with sobel filter dense adversarial noise secured layer framework for flower species recognition," *Scientific Reports*, vol. 15, no. 1, p. 12315, 2025.

[33] M. Toğaçar, B. Ergen, and Z. Cömert, "Classification of flower species by using features extracted from the intersection of feature selection methods in convolutional neural network models," *Measurement*, vol. 158, p. 107703, 2020.

[34] C. Kamann and C. Rother, "Benchmarking the robustness of semantic segmentation models with respect to common corruptions," *International journal of computer vision*, vol. 129, no. 2, pp. 462–483, 2021.

[35] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *2018 IEEE/CVF conference on computer vision and pattern recognition*, pp. 4510–4520, Ieee, 2018.

[36] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *2017 IEEE conference on computer vision and pattern recognition (CVPR)*, pp. 2261–2269, Ieee, 2017.

[37] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, "Inception-v4, inception-resnet and the impact of residual connections on learning," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 31, 2017.

[38] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.

[39] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2818–2826, 2016.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.

[41] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255, Ieee, 2009.

[42] I. Goodfellow, "Deep learning," 2016.

[43] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," *The journal of machine learning research*, vol. 15, no. 1, pp. 1929–1958, 2014.

[44] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[45] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[46] T. Panboonyuen, "Seeing isn't always believing: Analysis of grad-cam faithfulness and localization reliability in lung cancer ct classification," *arXiv preprint arXiv:2601.12826*, 2026.

[47] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks," in *2018 IEEE winter conference on applications of computer vision (WACV)*, pp. 839–847, IEEE, 2018.

[48] P.-J. Kindermans, S. Hooker, J. Adebayo, M. Alber, K. T. Schütt, S. Dähne, D. Erhan, and B. Kim, "The (un) reliability of saliency methods," in *Explainable AI: Interpreting, explaining and visualizing deep learning*, pp. 267–280, Springer, 2019.

[49] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu, "Score-cam: Score-weighted visual explanations for convolutional neural networks," in *2020 IEEE/CVF conference on computer vision and pattern recognition workshops (CVPRW)*, pp. 111–119, IEEE, 2020.

[50] P.-T. Jiang, C.-B. Zhang, Q. Hou, M.-M. Cheng, and Y. Wei, "Layercam: Exploring hierarchical class activation maps for localization," *IEEE transactions on image processing*, vol. 30, pp. 5875–5888, 2021.

[51] V. Petsiuk, A. Das, and K. Saenko, "Rise: Randomized input sampling for explanation of black-box models," *arXiv preprint arXiv:1806.07421*, 2018.

[52] A. Nastoska, B. Jancheska, M. Rizinski, and D. Trajanov, "Evaluating trustworthiness in ai: Risks, metrics, and applications across industries," *Electronics*, vol. 14, no. 13, p. 2717, 2025.

[53] M. Pal, H. N. Saha, and A. Chakrabarti, "The trust-aware xai (taxai) framework: a quantitative model for interpretable and reliable clinical ai systems," *Scientific Reports*, 2026.

[54] S. Schneider, E. Rusak, L. Eck, O. Bringmann, W. Brendel, and M. Bethge, "Improving robustness against common corruptions by covariate shift adaptation," *Advances in neural information processing systems*, vol. 33, pp. 11539–11551, 2020.

[55] F. Zhang, Q. Wang, and J. Zhao, "Ar2: Attention-guided repair for the robustness of cnns against common corruptions," *arXiv preprint arXiv:2507.06332*, 2025.

[56] T. Mahmud, K. Barua, A. Barua, N. Basnin, S. Das, M. S. Hossain, and K. Andersson, "Explainable ai for tomato leaf disease detection: Insights into model interpretability," in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, pp. 1–6, IEEE, 2023.

[57] B. Singh, R. P. Sharma, and S. Dey, "Interpretable plant leaf disease detection using attention-enhanced cnn," *arXiv preprint arXiv:2512.17864*, 2025.

[58] S. M. Shawon, F. L. Niha, O. H. Sinan, M. J. Abed, M. Al Qwaid, M. T. Sarker, and H. Zubair, "Integrating federated learning and explainable ai for plant disease detection: A systematic review of recent advances in precision agriculture," *International Journal of Computational Intelligence Systems*, 2026.

[59] G. Zheng, W. Ye, and A. Zhang, "Shortcutprobe: Probing prediction shortcuts for learning robust models," *arXiv preprint arXiv:2505.13910*, 2025.

[60] H. Baniecki, M. Chrabaszcz, A. Holzinger, B. Pfeifer, A. Saranti, and P. Biecek, "Be careful when evaluating explanations regarding ground truth," *arXiv preprint arXiv:2311.04813*, 2023.

[61] E. Kim, S. Kim, J. Lee, H. Kim, and S. Yoon, "Bridging the gap between classification and localization for weakly supervised object localization," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14238–14247, IEEE, 2022.